Draw forces on one chosen object

LEARNING TARGET · S5 I can draw every force on one object, with arrows to scale.

A · KEY VOCABULARY

Object	The receiver of every force in the diagram.
External	A force from outside the chosen boundary.
Length	Use one scale to show each force's size.
Net force	Sum of forces, not another interaction.

B · KEY EQUATIONS + UNITS

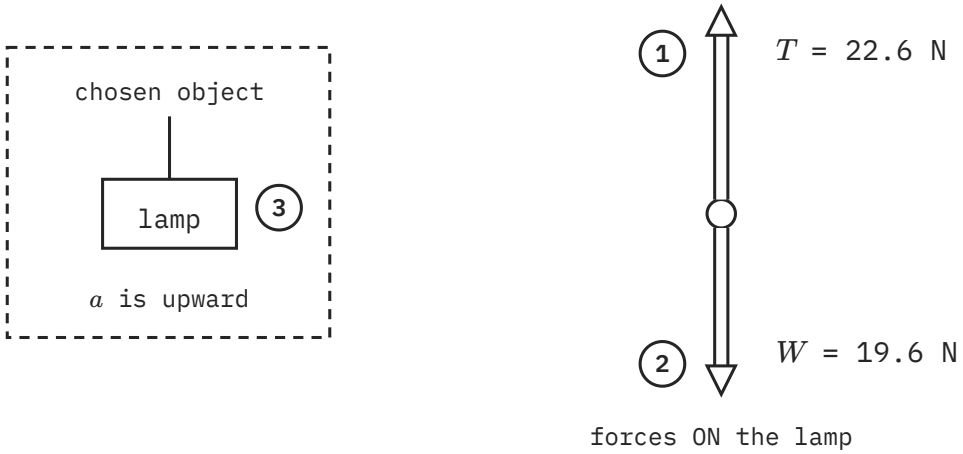
$$T - mg = ma_y$$

$$T = m(g + a_y)$$

\+ up
W Earth on lamp

C · COLOR IT · LAMP ACCELERATES UP

Forces on the lamp only.



- ☐ **1 · ORANGE** Tension: longer arrow
- ☐ **2 · ORANGE** Weight: shorter arrow
- ☐ **3 · PENCIL** Lamp and boundary

D · WORKED EXAMPLE

A rope lifts a **2 kg** lamp with upward acceleration **1.5 m/s²**. **Find the tension.** Use g = **9.8 m/s²**.

GIVEN	EQUATION	WORK	ANSWER
m = **2 kg** a_y = **+1.5 m/s²** g = **9.8 m/s²**	Up is +. $T - mg = ma_y$.	W = 2(9.8) = **19.6 N** T = 19.6 + 2(1.5)	T = **22.6 N** T and W are not a third-law pair.

E · THE 2026-XJ THREAD

DAY 15 OF 22 · NEW TOOL: OBJECT BOUNDARY

An XJ force diagram contains forces on XJ, not the forces XJ exerts on other bodies. Defining that boundary keeps the acceleration calculation consistent.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else